

Topological phonons

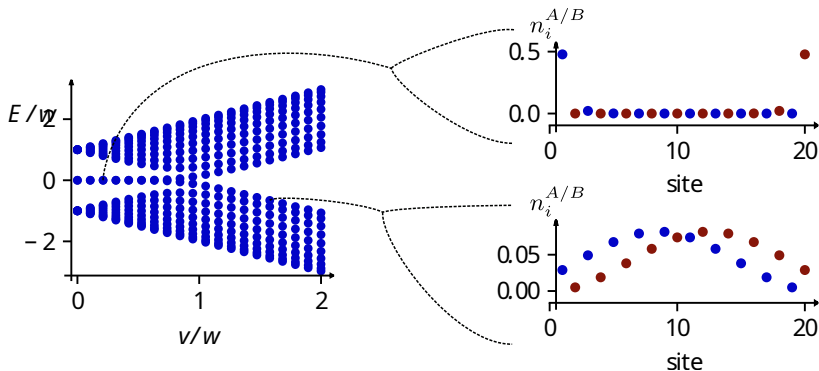
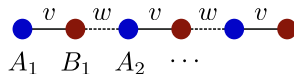
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SSH-model

Hamiltonian:

$$H_{\text{ssh}} = \sum_{i=1}^{N/2} \left[v a_i^\dagger b_i + w b_i^\dagger a_{i+1} \right] + \text{h.c.}$$

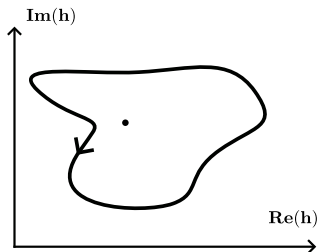
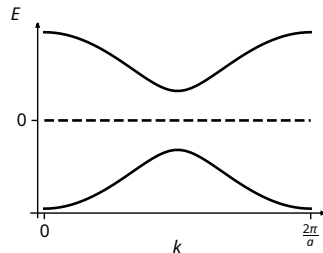


Classification and robustness

Phase determined by symmetries of the bulk:
bulk-boundary correspondence

Example: chiral symmetry

$$\rightarrow H = \begin{pmatrix} 0 & h(\vec{k}) \\ h^\dagger(\vec{k}) & 0 \end{pmatrix}$$

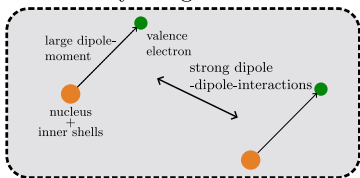


Topological phases can be distinguished
 by topological invariants

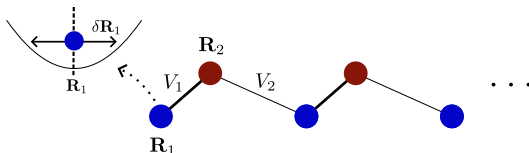
in 1D: winding number

Physical implementation

Rydberg atoms



preparing the atoms in the Rydberg states:



$$H = H_{\text{Trap}} + \sum_{i \neq j} V(\mathbf{R}_i - \mathbf{R}_j) \rightarrow H_{\text{eff}} = \sum_{i,j} U_{i,j} a_i^\dagger a_j$$

$$\text{for } i \neq j: U_{ij} = \frac{1}{m\omega} \frac{\partial^2 V(\mathbf{R}_i - \mathbf{R}_j)}{\partial \delta \mathbf{R}_i \partial \delta \mathbf{R}_j}$$

1D phonons

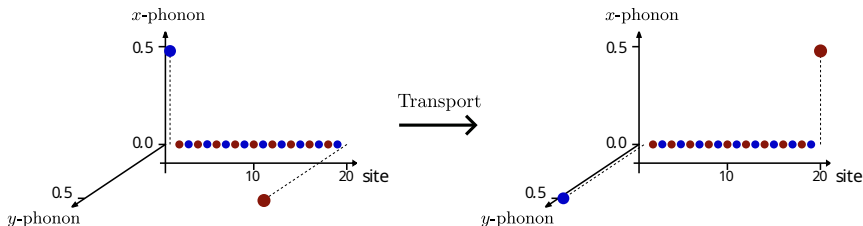
- $U_{i,i} = 0$
- can be mapped to the extended ssh model

2D phonons

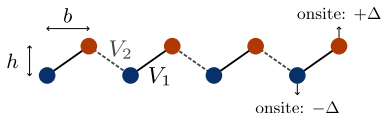
- $U_{i,i} = 0 \leftarrow$ only if phonon directions are decoupled
- two independent nearest neighbor ssh models

Rice-Mele scheme

Transport of phononic excitations



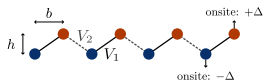
Rice-Mele scheme:



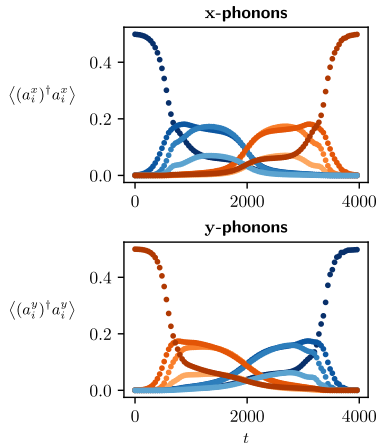
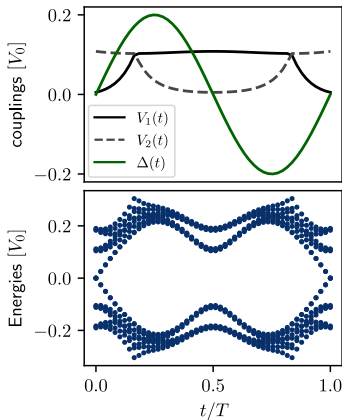
Adiabatic and periodic change
in parameters: $b(t)$, $h(t)$, $\Delta(t)$

¹S. Raupach et al., *Robust topological quantum state transfer with long-range interactions in rydberg arrays*, Jan. 27, 2026

Transfer results



A_1 B_1 ...



Model

- Go to higher dimensional lattices, e.g. transport on the closed surface of a 2D-lattice

- Spin-phonon coupling:

$$H = H_{\text{sp}} + H_{\text{sp-ph}}$$

- What effect has the spin-phonon coupling on transport performance?
- Is it possible to transport both spins and phonons simultaneously?

Thank you